

Budgeted Marginal Inference for Field-Level Risk Grading of Power-System Data

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Abstract—Power-system data are graded and released field by field, but inference risk often arises from combinations: individually uninformative fields can jointly reveal unit outputs, line flows, or load. We address this mismatch by deriving field-level risk from set-level inferability, measured by the out-of-sample accuracy of retrained attackers. The budgeted marginal risk of a field is its largest inferability gain over any background of at most K other fields, relative to data already public. This quantity identifies both a field’s worst-case contribution and a witness combination that explains its risk. To make the resulting audit tractable, an amortized attacker combines masked pre-training with a closed-form readout for each field set; retraining selected backgrounds then certifies lower bounds on field risk. Experiments cover ten targets across RTS-GMLC, the Australian National Electricity Market, PJM, and CAISO, with 124,425 exhaustively evaluated field sets. Single-field grading recovers only 13–17% of critical fields and leaves over 95% of minimal unsafe combinations intact. Scan–certify recovers up to 88% with no false positives against the retrained reference, while estimate-driven withholding breaks 94–99% of unsafe combinations with near-minimal withholding. Ablations identify the subset-specific readout as the main source of estimator accuracy. Increasing the background budget from two to three fields changes marginal risk little for most targets but reveals additional threshold-crossing combinations relevant to release decisions.

Index Terms—Power-system data, inference risk, data sensitivity, field combinations, marginal contribution, amortized inference.

I. INTRODUCTION

Power-system operators release forecasts, prices, reserve requirements, and operating statistics to support transparent and efficient operation. These records describe a shared physical system, so their information content is interdependent. A price series and a renewable forecast may each reveal little about a generating unit; together, they can help reconstruct the residual demand that determines its dispatch. A release decision made separately for each field can therefore overlook the information exposed by their combination.

This problem is particularly relevant to data classification and grading. Disclosure rules specify which operational fields are released and when [1], while classification standards call for aggregation and processing effects to be considered [2]. The audit nevertheless needs a usable output at the field level: which field poses a risk, how much risk it adds, and which other fields make that risk material. The inference and aggregation problem has long been recognized in database security [3], [4]; in power systems, even public prices can reveal grid topology [5]. Translating this set-dependent leakage into an actionable field-level assessment remains the central challenge.

Existing scores address parts of the problem. Correlation and single-field prediction measure what one field reveals

alone. Feature-attribution methods distribute predictive value or evaluate a field in a particular fitted model. Inference-graph methods propagate sensitivity through relations between fields [6]. For a release audit, however, the relevant question is whether a field can substantially improve an adversary’s inference in *any admissible background*. A field may have low individual predictability, little average attribution, and yet be decisive in one combination.

We take the predictive accuracy of attackers trained on a field set as the starting point. A field’s *budgeted marginal inference risk* is its largest increase in this accuracy over backgrounds containing at most K other fields, relative to the public baseline. The score adapts the marginal contribution index [7] to a bounded-background release audit. Its maximizing background supplies an explanation: the risk score is accompanied by the fields with which the contribution arises. Threshold-crossing combinations then connect the assessment to field flagging and withholding.

The computational difficulty is evaluating many such backgrounds. Even 35 candidate fields produce 7,175 nonempty sets of size at most three. We therefore share a nonlinear representation across sets and fit a closed-form readout separately for each query. This amortized attacker scans the field combinations; dedicated retraining of a few selected backgrounds verifies the largest contributions. Fig. 1 summarizes the resulting audit.

The contributions are threefold:

- A field-level risk measure that retains the combination structure of set-level inferability. We establish its safety-margin interpretation and its connection to minimal unsafe sets under monotonicity, and use witness backgrounds to explain field risk.
- An amortized estimator based on masked pre-training and subset-specific readouts, together with a scan–certify procedure that recovers retrained lower bounds from a small number of backgrounds.
- An evaluation over four power-system data sets, ten targets, and 124,425 field sets. The experiments measure missed combination risk, compare grading and withholding strategies, isolate the estimator’s effective components, and assess robustness and background-budget coverage.

II. RELATED WORK

a) Inference and aggregation in data release: That individually innocuous data items can jointly reveal protected information is an old observation in database security, where it is known as the inference and aggregation problem [3],

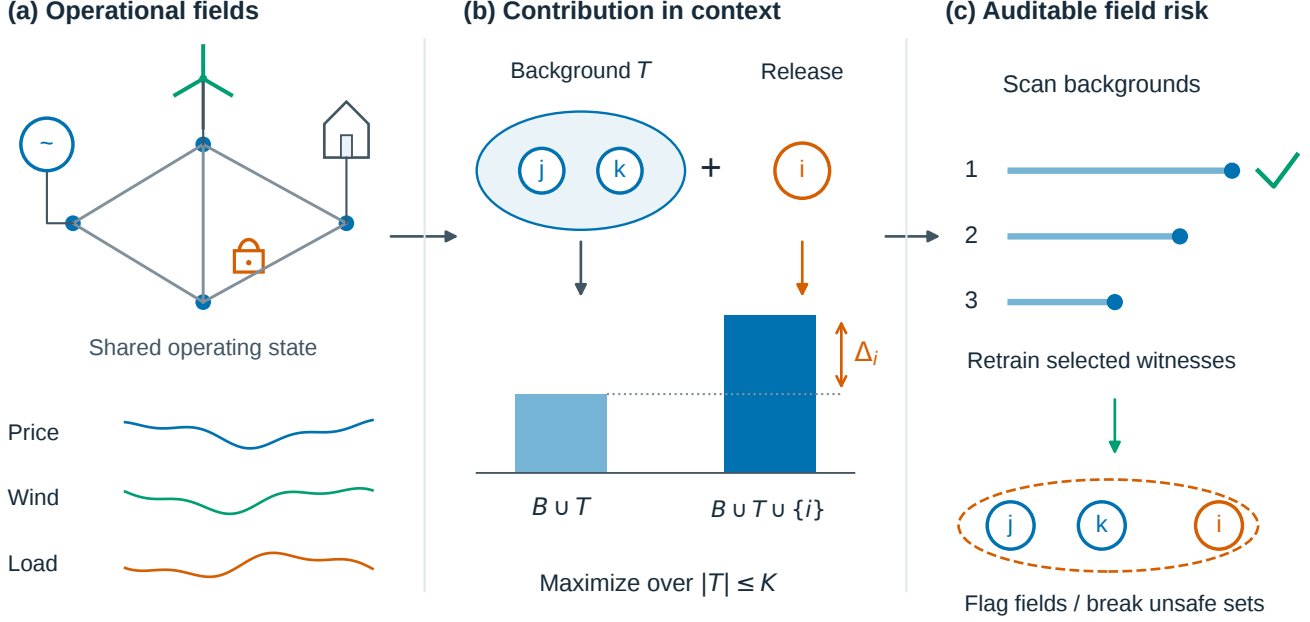


Fig. 1. From operational data to a field-level audit. The grid and traces illustrate the shared physical origin of the released fields. A field is evaluated in combination with a background T relative to the public baseline B ; the largest gain defines its risk and supplies a witness. An amortized scan selects backgrounds for retrained assessment, while unsafe combinations support withholding decisions. The grid and combinations are schematic.

[4]; the privacy literature supplies the best-known illustration, three demographic attributes that jointly single out most individuals [8]. In power systems, public market data have been shown to expose non-public physical information, for example grid topology from locational marginal prices [5]. Privacy mechanisms for metering data [9], [10] and formal models such as differential privacy [11] or k -anonymity [12] bound what can be learned about an individual record after a release mechanism is applied. They answer a different question from the one faced by a market operator, who must decide which aggregate operational *fields* may be published at all.

b) Data sensitivity assessment under inference leakage: Data classification and grading is performed field by field in practice, and the Chinese national standard explicitly asks that aggregation and processing effects be considered [2], without prescribing how. The closest work to ours is IGNN [6], which introduced data inference leakage into sensitivity assessment for cyber-physical power grids: fields are nodes of an inference graph whose edges encode physical or statistical inference relations, and sensitivity is propagated along these edges by a graph attention network from an intrinsic (expert-assigned) score. This established that the risk of a field depends on its inference relations to sensitive data. Two questions remain open in this line of work. First, the risk score is calibrated against expert labels, while its ultimate validation is an attack experiment in which a model is trained on a group of fields; the attack outcome itself is not the definition. Second, an edge connects two fields, whereas inference frequently requires several fields jointly, with none of them informative alone. We take the attack outcome on a field *set* as the definition of

risk and derive the field-level quantity from it.

c) Feature importance: Turning set-level values into per-feature scores is the subject of feature attribution. Shapley-based measures [13], [14], [15], [16] average a feature's marginal contribution over all contexts, providing an allocation of predictive value that can dilute the credit of redundant features [17], [7]. Leave-one-covariate-out (LOCO) [18] and permutation importance [19], [20], [21] evaluate a single context, the full feature set, so a feature with a near-substitute receives almost no importance. The Marginal Contribution Index (MCI) [7] takes the *maximum* marginal contribution over contexts, is characterized axiomatically, and is immune to dilution. Worst-case assessment also motivates min-entropy and gain-function approaches in quantitative information flow [22], [23]: an average over contexts is not a guarantee. Our field-level risk is a budget-constrained MCI evaluated relative to already-public data.

d) Fast estimation of many refitted models: Computing the performance of a model refitted on many feature subsets is expensive, which has motivated lazy [24] and warm-started [25] estimators of variable importance, amortized explanation models [26], [27], and conditional generative models trained under random masks [28], [29], [30], [31]. Closed-form ridge heads on frozen features are the standard fast-adaptation device in meta-learning [32], [33], and the benefit of sharing a representation across tasks is well understood [34], [35], [36]. We combine these ingredients into an estimator of set-level inferability and use it inside a scan-certify loop to recover retrained lower bounds on the largest field contributions.

III. FROM SET-LEVEL INFERENCE TO FIELD-LEVEL RISK

A. Parties, field roles and threat model

The **auditor** is the data owner. It holds the complete history of all operational fields, including those it regards as sensitive, and grades fields offline before release. The **adversary** obtains published fields and is assumed to hold historical auxiliary labels of the sensitive target, from which it trains a supervised inference model; risk is therefore measured as out-of-sample *supervised* inferability. The adversary may combine the field under review with other fields it already holds.

Field roles are fixed by disclosure rules, not by data analysis. Let G be the candidate fields that the rules allow to be published, $B \subseteq G$ a *public baseline* that is already released ($B = \emptyset$ unless stated otherwise), $H = G \setminus B$ the fields under review, and Y_y a sensitive target. Section V-A instantiates these roles for four data sets from the categories and release timings of the Chinese electricity-market disclosure rules [1]. An *ex-ante* estimate of the *same physical quantity* as a target (e.g. a load forecast when the target is metered load) is removed from G , to focus the audit on inference across distinct operational quantities.

B. Set-level inferability

Definition 1 (Population inferability).

$$V_y^*(S) = 1 - \inf_f \mathbb{E}[(Y_y - f(X_S))^2] / \text{Var}(Y_y),$$

the infimum taken over measurable f . It satisfies $V_y^*(\emptyset) = 0$, $0 \leq V_y^* \leq 1$ and is monotone in S .

Definition 2 (Empirical inferability). *The data are split into training and test rows, with a validation part inside training. For an attacker family $\mathcal{H}$, every $h \in \mathcal{H}$ is trained on X_S only; the attacker and its training epoch are selected on validation, and the test R^2 is reported: $V_y^{\text{ret}}(S) = R_{\text{test}}^2(\hat{h}_S)$ with $\hat{h}_S = \arg \max_{h \in \mathcal{H}, \text{epoch}} R_{\text{val}}^2(h; S)$.*

Finite samples, optimization and selection noise make V_y^{ret} slightly non-monotone. An adversary holding S can train on any subset of S and keep the best, so the operational evaluation allows subset selection

$$V_y(S) = V_y^{\text{ret}}(\hat{T}_S), \quad \hat{T}_S = \arg \max_{T \subseteq S} R_{\text{val}}^2(T), \quad (1)$$

where the subset is chosen on validation and the value is read on test. We write V_y for the resulting retrained reference throughout. It supplies the set values against which field-risk estimates and release decisions are evaluated.

C. Budgeted marginal risk

A release audit must account for joint inference, redundant substitutes, and the information already available to an adversary. We therefore evaluate each field in its most informative admissible background, with the background size bounded by K . Appendix B compares these requirements with existing field scores.

Definition 3 (Budgeted marginal inference risk). *For target y , public baseline B , fields under review $H = G \setminus B$ and budget K ,*

$$M_{i \rightarrow y}^{(K)} = \max_{T \subseteq H \setminus \{i\}, |T| \leq K} \Delta_i(B \cup T), \quad (2)$$

$$\Delta_i(A) = V_y(A \cup \{i\}) - V_y(A).$$

Any maximizer T^* is a witness background.

$M_{i \rightarrow y}^{(0)} = V_y(B \cup \{i\}) - V_y(B)$ is single-field risk, the quantity current practice grades on; the *escalation* $M_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(0)} \geq 0$ is the risk that appears only in combination; the *gain matrix* $V_y(\{i, j\}) - V_y(\{j\})$ lists the increment of i over every single background.

Proposition 1 (Relation to MCI). *For monotone V_y , let $v_B(T) := V_y(B \cup T) - V_y(B)$ for $T \subseteq H$. Then v_B is monotone, non-negative and $v_B(\emptyset) = 0$; $M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} [v_B(T \cup \{i\}) - v_B(T)]$ is the budget-constrained MCI of v_B ; for $K \geq |H| - 1$ and $B = \emptyset$ it is the MCI of V_y .*

Proof. Monotonicity of V_y gives monotonicity of v_B , and $v_B(\emptyset) = 0$ by construction; the rest is substitution. $\square$

The axiomatic properties of MCI for monotone evaluation functions [7]—dummy, symmetry, the self-contribution lower bound $M_{i \rightarrow y}^{(K)} \geq M_{i \rightarrow y}^{(0)}$ and invariance to duplicated features—therefore transfer to this setting. The finite budget gives the following interpretation for a release audit.

D. From marginal risk to unsafe combinations

Theorem 1 (Threshold-free safety margin). *Fix i, y, B, K . A number $m \geq 0$ satisfies “for every threshold τ and every background $|T| \leq K$: $V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau$ ” if and only if $m \geq M_{i \rightarrow y}^{(K)}$. For a fixed τ , the smallest m with this property can be strictly smaller than $M_{i \rightarrow y}^{(K)}$.*

Proof. If $m \geq M_{i \rightarrow y}^{(K)}$ then $V_y(B \cup T \cup \{i\}) = V_y(B \cup T) + \Delta_i(B \cup T) \leq (\tau - m) + m$. Conversely, for any T choose $\tau = V_y(B \cup T) + m$; the property gives $\Delta_i(B \cup T) \leq m$. Strictness: with $B = \emptyset$, $K = 1$, $V_y(\{i\}) = 0.1$, $V_y(\{j\}) = 0.5$, $V_y(\{i, j\}) = 0.9$ one has $M_{i \rightarrow y}^{(1)} = 0.4$, yet at $\tau = 0.95$ the necessary margin is 0. $\square$

Thus $M_{i \rightarrow y}^{(K)}$ is the answer to “by how much can publishing i raise the adversary’s accuracy, whatever it already holds within budget”—a property of the field that does not depend on the threshold later chosen for grading.

Definition 4 (Criticality and minimal unsafe sets). *Under baseline B , $E \subseteq H$ is a τ -minimal unsafe set (MUS) if $V_y(B \cup E) > \tau$ and $V_y(B \cup E') \leq \tau$ for every $E' \subsetneq E$. Field i is K -background τ -critical if some T with $|T| \leq K$, $i \notin T$, has $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$.*

Theorem 2 (Criticality $\iff$ small MUS). *If V_y is monotone and $V_y(B) \leq \tau$, field i is K -background τ -critical if and only if it belongs to a MUS of size at most $K + 1$.*

Proof. ($\Leftarrow$) For a MUS $E \ni i$ with $|E| \leq K + 1$ take $T = E \setminus \{i\}$. ($\Rightarrow$) If $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$, take

an inclusion-minimal unsafe $E \subseteq T \cup \{i\}$; it is a MUS of size $\leq K + 1$. If $i \notin E$ then $E \subseteq T$ and monotonicity gives $V_y(B \cup T) \geq V_y(B \cup E) > \tau$, a contradiction. $\square$

Theorem 2 makes the field-level flag a faithful projection of the set-level danger structure: flagging the τ -critical fields misses no field that takes part in a dangerous combination of at most $K + 1$ fields, and a threshold-crossing background identifies an unsafe combination. The same MUS list gives the smallest set of fields whose withholding breaks every dangerous small combination (a minimum hitting set), which we use for withholding.

Proposition 2 (Closed form at $K = 1$). *With $I_{ij}^B = V_y(B \cup \{i, j\}) - V_y(B \cup \{i\}) - V_y(B \cup \{j\}) + V_y(B)$, $M_{i \rightarrow y}^{(1)} = M_{i \rightarrow y}^{(0)} + \max(0, \max_j I_{ij}^B)$.*

Hence a field escalates at $K = 1$ exactly when it is strictly super-additive with some partner.

E. Background budget

$M_{i \rightarrow y}^{(K)}$ is non-decreasing in K , so K trades coverage against the cost of evaluating all sets of size $\leq K + 1$. We use $K = 2$, for three reasons. The threat model is an adversary holding a handful of other published series, not the whole catalogue. Empirically, high-order interactions are rare in engineered systems (the hierarchy principle of factorial experiments [37]), and models restricted to pairwise interactions are close to unrestricted ones on tabular data [38]. Finally, we measure saturation directly on every data set (Section VI-D). Two caveats apply. There is no general theorem—a parity function of $K + 2$ fields defeats any fixed K —and saturation of the marginal gain does not imply that no larger set crosses the threshold. Section VI-D shows that both matter in practice: $K = 2$ captures most of the marginal risk, but withholding decisions, which must break every dangerous combination, are better computed at $K = 3$.

IV. AN AMORTIZED ATTACKER FOR SET-LEVEL INFERRABILITY

A. Why an estimator is needed

Evaluating $M_{i \rightarrow y}^{(K)}$ for all fields at $K = 2$ requires V_y on every set of size at most three; with 35 candidates this is 7,175 sets per target, each requiring a family of attackers to be trained. The table has to be recomputed whenever the public baseline grows, a target is redefined, or a what-if release is examined. We therefore build one model that answers $V_y(S)$ for any S quickly (Fig. 2), and use selected retrained backgrounds to assess the scan (Section IV-F).

B. Masked pre-training

The adversary does not know in advance which fields it will obtain. We model this by pre-training one network under random visibility masks. A mask $m \in \{0, 1\}^p$ is drawn by first sampling the number of visible fields uniformly and then a uniform subset of that size; the network $f_\theta(x \odot m, m)$ receives the masked values and the mask itself, so that a hidden field is

distinguishable from a zero value. The main estimator predicts the designated sensitive targets from the visible candidates. An extension using *masked reconstruction* predicts every hidden column of the auditor’s data from the visible ones [30], [31]; it involves no designated target, so one pre-training serves any later definition of targets and candidates. The last hidden layer (256 units) is kept as a frozen feature map $\phi_\theta(x \odot m, m)$. Early stopping uses the validation rows only.

C. Subset-specific closed-form readout

For a query (S, y) with mask m_S and seed r , the feature block is

$$Z_r(S) = [x_S, x_S^2, \phi_{\theta_r}(x_S, m_S)], \quad x_S = x \odot m_S, \quad (3)$$

standardized with training statistics. A ridge readout

$$\beta_r(S) = (Z_r^\top Z_r + \lambda I)^{-1} Z_r^\top y \quad (4)$$

is solved in closed form on the training rows, with λ chosen by five-fold cross-validation inside the training rows. The predictions of $R = 3$ independently pre-trained backbones are averaged, $\hat{y} = R^{-1} \sum_r Z_r^{\text{test}} \beta_r(S)$, and $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y})$, followed by the monotone closure. Two numerical safeguards matter on real data: a floor of 10^{-3} on feature standard deviations, and clipping test features to their training range. Without them, units that are almost constant on the training rows (inactive ReLUs) are amplified on test rows, a few predictions explode and $\hat{V}$ collapses to zero for those sets; since $M_{i \rightarrow y}^{(K)}$ is a maximum over differences, such collapses turn directly into inflated risk (Section VI-C). Pre-training, readout fitting, and the choice of λ use training and validation data.

D. Role of the subset-specific readout

A cheaper alternative lets the network output the prediction directly, $\hat{y} = f_\theta(x \odot m_S, m_S)$, and reads $\hat{V}$ off it. One set of output weights then serves every conditioning pattern, which can limit adaptation to the particular information available in a query. Solving the readout per subset keeps the expensive representation shared and the combination rule specific.

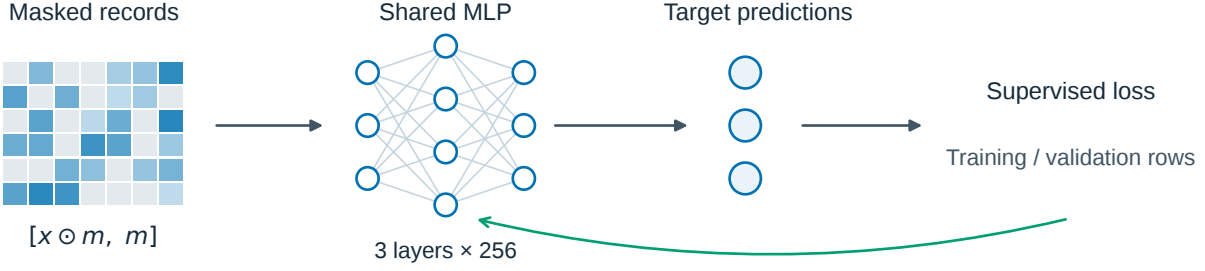
E. Error propagation

Write the estimate as $\hat{V}(S) = V(S) + b + \varepsilon(S)$, where b is a bias common to the sets of interest and $\varepsilon(S)$ the set-specific part. Since a marginal is a difference of two nested sets, $\hat{\Delta}_i(T) - \Delta_i(T) = \varepsilon(T \cup \{i\}) - \varepsilon(T)$ and

$$|\hat{M}_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(K)}| \leq \max_{|T| \leq K} |\varepsilon(T \cup \{i\}) - \varepsilon(T)|, \quad (5)$$

independently of b . A readout on a frozen representation typically underestimates the fully trained attacker by a similar amount on nested sets, so a shared component of that error can cancel. Thus bias in V need not translate into equally large error in M . This motivates evaluating marginal error alongside set-value error.

(a) Learn a conditional representation once



(b) Adapt the readout to each field set

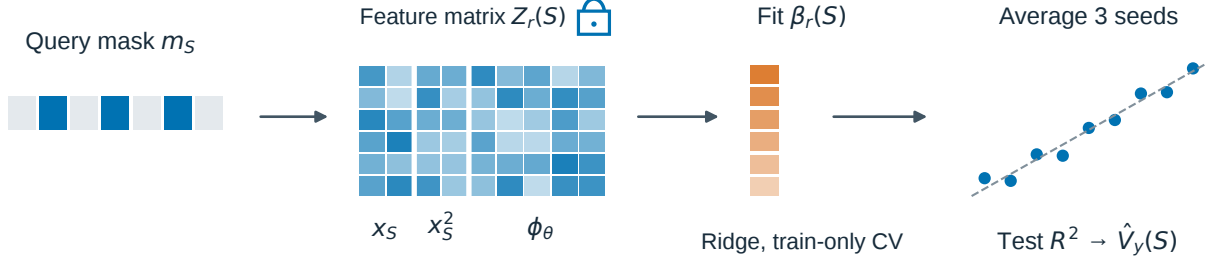


Fig. 2. The amortized attacker separates shared representation learning from adaptation to each field set. (a) Random visibility masks train a target-supervised backbone. (b) Each query concatenates masked values, their squares, and frozen hidden features; a separate ridge readout is fitted using training rows, and predictions are averaged across three seeds. The matrices, network nodes, and prediction marks are schematic. All-column reconstruction is evaluated separately as an extension.

F. Scan and certify

The estimator is used to *scan*: it produces $\hat{M}^{(K)}$ for every field together with a ranked list of candidate backgrounds. For *certification*, dedicated attackers are retrained on the top- k backgrounds only, and $L_{i \rightarrow y}^{(K)} = \max_{T \in \text{top-}k} \Delta_i(B \cup T) \leq M_{i \rightarrow y}^{(K)}$ is a lower bound computed entirely from retrained attackers. The ratio $\sum_i L / \sum_i M$ measures how much of the true risk the scan recovers at a certification cost of k retrained backgrounds per field instead of all $\binom{|H|-1}{\leq K}$.

V. EXPERIMENTAL SETUP

A. Data sets and rule-based field roles

Table I summarizes the four data sets and ten sensitive targets. Candidate and target roles follow the categories and release timings used in the disclosure-based audit [1]: candidates include forecasts, prices, and operating requirements; targets include unit outputs, an internal line flow, and actual operating quantities. These roles define a common evaluation setting across data sets. Release timing determines the field roles, while the train/test splits are random.

RTS-GMLC. The IEEE Reliability Test System 2019 update [39] provides a 73-bus network with one year of realistic hourly load, wind, solar and hydro series. We clear an hourly DC optimal power flow with piecewise-linear offers built from the published heat-rate curves (8,784 hours, 26.7% congested). Candidates are the 23 fields a market would publish early

(zonal load forecasts, renewable and hydro forecasts, reserve requirements, system energy price and nine nodal prices, three congestion indicators); targets are the outputs of a coal and a combined-cycle unit and the hourly flow on internal line C35.

NEM. One year (2024) of the Australian National Electricity Market [40], averaged to hours. Candidates are regional demand, available capacity, semi-scheduled wind and solar forecasts, regional prices and interconnector flows (30 fields); targets are the actual outputs of a coal unit (BW01), a hydro unit (TUMUT3) and a gas combined-cycle unit (PPCCGT). Unit output is public in the NEM but is specific information under the Chinese rules; this provides measured unit-output targets for the specified audit setting.

PJM and CAISO. System-level hourly data for one year. Targets are the actual quantities released on the next day: PJM metered load, total generation and net actual interchange, and CAISO actual load. Candidates are forecasts, day-ahead and real-time prices and their components, ancillary-service requirements and cleared volumes, and schedules (22–35 fields). Fields released on the same day as the target (fuel-mix generation, actual renewable output, other balancing areas' actual load) are excluded, as are the load forecasts of the target area, which predict the same physical quantity as the target (Section III). Related but distinct quantities (e.g. sub-area forecasts for CAISO load) are kept; removing them is analysed as a sensitivity case.

TABLE I
DATA SETS. n : TRAIN/TEST ROWS; p : CANDIDATE FIELDS; SETS: FIELD SETS WITH EXACT THREE-ATTACKER GROUND TRUTH.

Data set	Sensitive targets	n	p	sets	max size
RTS-GMLC	Unit 223, unit 221, line C35	6149/2635	23	10,902	4
NEM 2024	Units BW01, TUMUT3, PPCCGT	6149/2635	30	31,930	4
PJM (load)	Metered load	4589/1967	22	9,108	4
PJM (gen./interch.)	Total generation, net interchange	4589/1967	24	12,950	4
CAISO	Actual load	6105/2617	35	59,535	4

B. Ground truth

All variables are standardized with training statistics; 70/30 random train/test splits are used, with 15% of the training rows as validation. The attacker family has three members: a single-target multilayer perceptron (two hidden layers of 128 units, dropout 0.15, Adam, early stopping on validation), a multi-target perceptron trained on all targets of a data set, and histogram gradient boosting with two configurations. For every set of up to four fields, every member is trained on that set alone; the member and epoch are chosen on validation and the test R^2 is reported, followed by the monotone closure (1). This amounts to 124,425 field sets and several hundred thousand trained models. The amortized estimator is fitted without these retrained reference values.

C. Baselines

Field scores compared with $M^{(2)}$ are: Pearson r^2 , squared Spearman correlation, mutual information [41], distance correlation [42], single-field retraining $M^{(0)}$ (current practice), LOCO [18], permutation importance [19], SAGE with marginal imputation [15], and an inference-graph score that follows the propagation idea of [6] without its expert inputs: edge weights are the single-field predictability of one field from another (gradient boosting, validation R^2), the target node carries risk one, and risk is propagated over three hops with decay 0.5. All baseline scores are computed on training rows only.

Estimators compared with ours are: linear and quadratic readouts without a backbone, an in-set quadratic dictionary (values, squares and pairwise products of the fields in S), the shared output head of the same backbone, an untrained (randomly initialized) backbone with the same readout, and warm-start fine-tuning of the whole network for each set [25].

D. Metrics

For set values: mean absolute error of $\hat{V}$. For field risk: mean absolute error and bias of $\hat{M}^{(2)}$, Spearman correlation across fields, and the certified ratio $\sum_i L_{i \rightarrow y}^{(2)} / \sum_i M_{i \rightarrow y}^{(2)}$ when k backgrounds per field are retrained. For decisions at thresholds $\tau \in \{0.5, 0.7\}$: recall and precision of critical fields; the number of fields withheld and the number of minimal unsafe sets of size ≤ 3 left unbroken.

VI. EXPERIMENTAL RESULTS

A. Combination risk missed by single-field grading

Table II and Fig. 3 report, for each of the ten targets, how many candidate fields are unsafe on their own and how many become critical only in combination. The results show a consistent gap between individual predictability and combination risk.

Most critical fields are invisible to single-field grading. Of the 264 field–target pairs, 256 are critical at $\tau = 0.5$ with at most two background fields, and 223 of these are safe on their own; at $\tau = 0.7$ the figures are 138 and 113. For four targets (NEM BW01 and TUMUT3, PJM load and PJM net interchange) no candidate exceeds $\tau = 0.5$ alone, yet 22 to 30 candidates take part in a dangerous combination; at $\tau = 0.7$ the same holds for RTS line C35, where 21 of 23 fields are critical and none is unsafe alone. Withholding every field that is unsafe alone breaks almost none of the minimal unsafe sets (column “left”): 1,766 of 1,799 at $\tau = 0.5$ and 534 of 559 at $\tau = 0.7$ remain.

Where single fields do leak, combinations still dominate. Some targets are exposed by a single close proxy: RTS unit outputs are almost determined by a nodal price ($V_1 = 0.99$), and PJM total generation by the system load forecast (0.97). For these, $M^{(0)}$ already flags the proxy, and the remaining risk is again combinatorial (e.g. 22 of 24 PJM fields are combination-critical). Removing such near-proxies from the candidate pool (Section V-A) lowers the best single field to 0.42 for both PJM total generation and CAISO load but leaves every remaining field combination-critical at $\tau = 0.5$ (22/22 and 26/26), so the conclusion does not depend on where the proxy boundary is drawn.

The witnesses are physically meaningful. Fig. 4 shows the field profile of RTS line C35 and the gain matrix of NEM unit PPCCGT. The largest escalations come from combinations that reconstruct a residual quantity: the SP15 day-ahead energy price rises from $M^{(0)} = 0.14$ to $M^{(2)} = 0.51$ for CAISO load with the NP15 wind and SP15 solar forecasts as witnesses (consistent with a net-load mechanism); the South Australian wind forecast rises from 0.19 to 0.60 for PPCCGT together with the regional available capacity and solar forecast (consistent with residual-demand-driven dispatch); the price at RTS bus 303 rises from 0.36 to 0.65 for the C35 flow together with a zonal load forecast and the congestion indicator of the neighbouring line C6. The gain matrix also shows that the same field is harmless next to some partners and decisive next

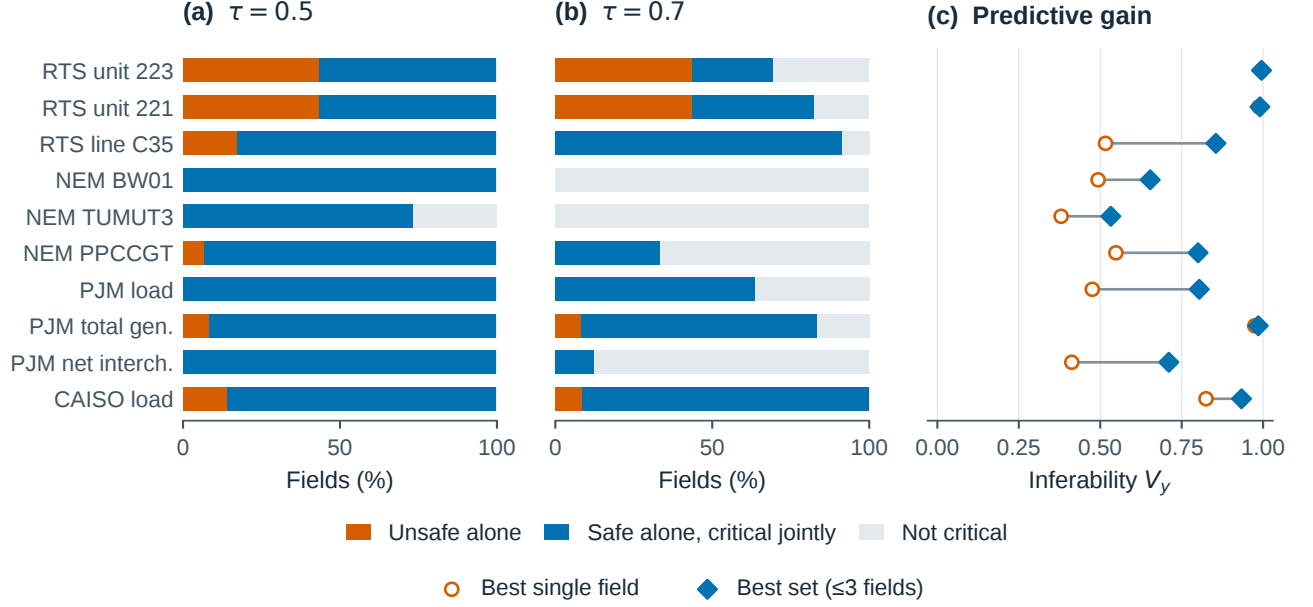


Fig. 3. Share of fields that are unsafe alone, safe alone but critical with at most two background fields, or not critical at (a) $\tau = 0.5$ and (b) $\tau = 0.7$. (c) The best single-field inferability compared with the best set of at most three fields. Rows share the same target order across panels.

TABLE II

COMBINATION RISK THAT SINGLE-FIELD GRADING MISSES. V_1 : BEST SINGLE FIELD; $V_{\leq 3}$: BEST SET OF AT MOST THREE FIELDS; ALONE: FIELDS WITH $V(\{i\}) > \tau$; COMB.: FIELDS SAFE ALONE BUT τ -CRITICAL WITH AT MOST TWO BACKGROUND FIELDS; MUS: MINIMAL UNSAFE SETS OF SIZE ≤ 3 ; LEFT: MUS NOT BROKEN AFTER WITHHOLDING EVERY FIELD THAT IS UNSAFE ALONE; HIT: MINIMUM NUMBER OF FIELDS WHOSE WITHHOLDING BREAKS EVERY MUS.

Target	p	V_1	$V_{\leq 3}$	alone	$\tau = 0.5$				alone	$\tau = 0.7$			
					comb.	MUS	left	hit		comb.	MUS	left	hit
RTS unit 223	23	1.00	1.00	10	13	60	50	15	10	6	16	6	11
RTS unit 221	23	0.99	0.99	10	13	52	42	15	10	9	34	24	13
RTS line C35	23	0.52	0.86	4	19	123	119	13	0	21	82	82	7
NEM BW01	30	0.49	0.65	0	30	266	266	4	0	0	0	0	0
NEM TUMUT3	30	0.38	0.53	0	22	23	23	1	0	0	0	0	0
NEM PPCCGT	30	0.55	0.80	2	28	215	213	6	0	10	8	8	1
PJM load	22	0.48	0.80	0	22	237	237	9	0	14	44	44	3
PJM total gen.	24	0.97	0.99	2	22	179	177	11	2	18	39	37	5
PJM net interch.	24	0.41	0.71	0	24	102	102	3	0	3	1	1	1
CAISO load	35	0.82	0.93	5	30	542	537	17	3	32	335	332	10

to others, which is what a grade that ignores the background cannot represent.

B. Field grading and withholding

a) *Why the scores differ:* Fig. 5 uses synthetic data with a known structure, $Y = aAB + bP + \sigma\epsilon$, where A and B act only jointly, C_1 and C_2 are two noisy copies of P , D is a weak copy of P and N is noise; all 63 sets are retrained. Correlation, single-field retraining and pairwise inference-graph propagation give A and B a score of zero although $\{A, B\}$ reaches 0.54. LOCO gives C_1 and C_2 almost zero because each substitutes for the other, and LOCO, permutation importance and SAGE all give D zero because the full model relies on C_1, C_2 instead—yet $\{A, B, D\}$ crosses $\tau = 0.7$ while $\{A, B\}$ does not, so D is critical. SAGE splits credit within both pairs. $M^{(2)}$ assigns A, B, C_1, C_2 their full worst-case

contribution and gives D a non-zero risk with the witness $\{A, B\}$. The example separates three effects: joint information from A, B , substitution between C_1, C_2 , and the context-specific contribution of D .

b) *Flagging and withholding:* We next compare field-flagging rules (Table III, Fig. 6a). The single-field rule recovers 13% of the critical fields at $\tau = 0.5$ and 17% at $\tau = 0.7$. The scan on amortized estimates recovers 93% and 83%; with certification, which only accepts a field after dedicated attackers retrained on one of its three best-scanned backgrounds confirm the threshold crossing, precision against the retrained reference is one and recall is 88% ($\tau = 0.5, \delta = 0$) and 80% ($\tau = 0.7, \delta = 0.05$). The margin δ compensates for the common-mode underestimation of the estimator (Section VI-C), which matters near high thresholds.

For withholding, the relevant question is how many fields

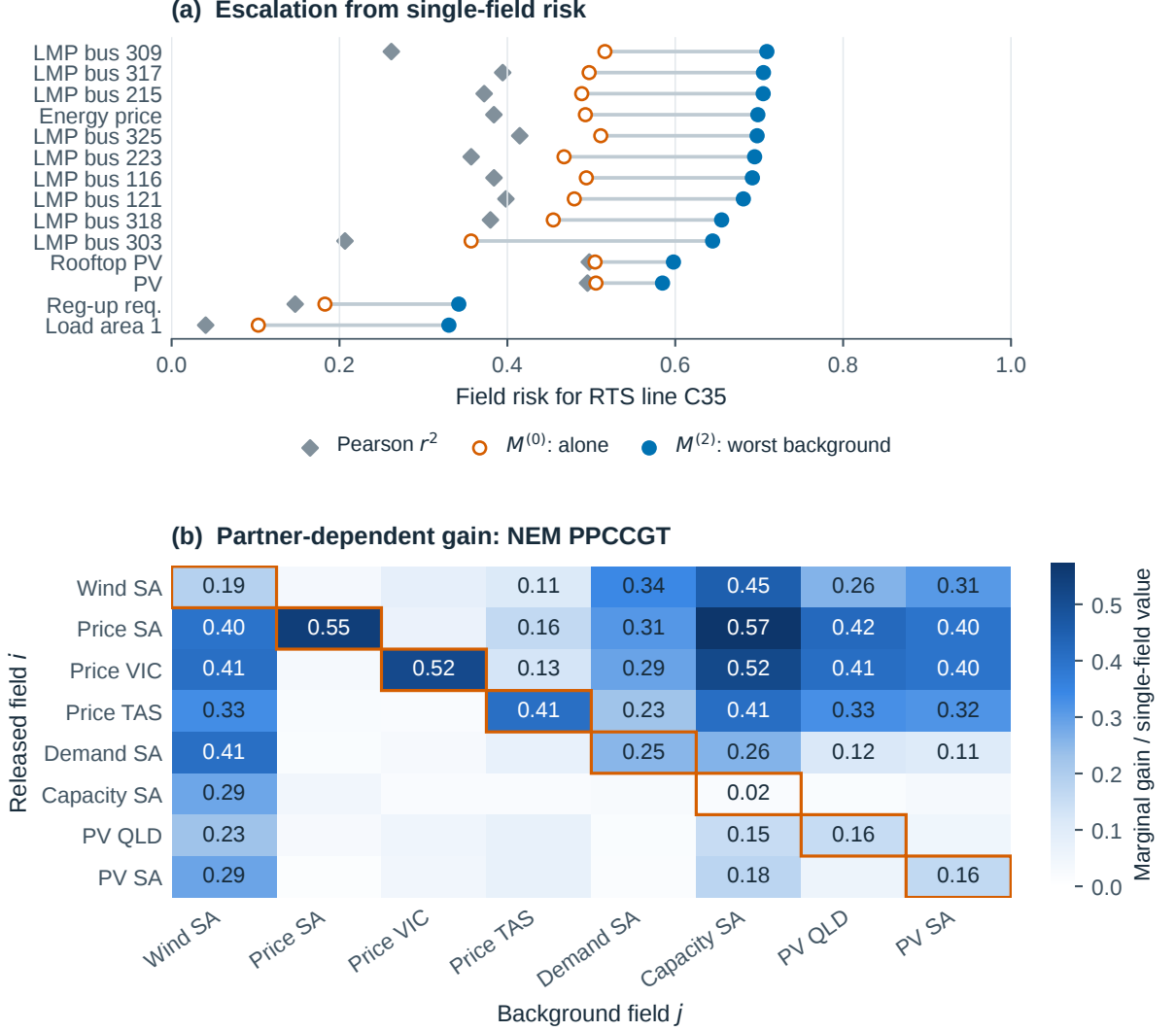


Fig. 4. Context explains field risk. (a) The 14 largest $M^{(2)}$ values for RTS line C35, compared with single-field risk and Pearson r^2 . (b) An eight-field view of the gain matrix for NEM PPCCGT: off-diagonal entry (i, j) is $V(\{i, j\}) - V(\{j\})$; the orange-outlined diagonal gives $V(\{i\})$. Off-diagonal values below 0.10 are omitted from the annotations.

must be withheld to break every dangerous combination. Withholding what single-field grading flags leaves 98% ($\tau = 0.5$) and 96% ($\tau = 0.7$) of the minimal unsafe sets intact. Two strategies that select fields from amortized estimates only—a minimum hitting set of the estimated minimal unsafe sets, and a greedy rule that withholds the field with the largest $\hat{M}^{(2)}$ and recomputes $\hat{M}^{(2)}$ on the remaining fields—leave 1.3–5.8% intact while withholding at most 1.2 fields more than the optimum computed from the ground truth (Fig. 6c). The greedy rule exploits the relative nature of $M_{i \rightarrow y}^{(K)}$: once a field is withheld, the risk of the others changes, and $M_{i \rightarrow y}^{(K)}$ is recomputed with respect to what remains.

c) *Ranking*: Table IV compares the scores as rankings. At $\tau = 0.7$, $\hat{M}^{(2)}$ has the highest R-precision (0.903, against 0.866 for mutual information, 0.830 for the inference graph, 0.803 for single-field retraining, 0.769 for SAGE and 0.727

for LOCO), and withholding fields in the order of exact or certified $M^{(2)}$ exceeds the optimum by only 0.4 fields. At $\tau = 0.5$ almost every field is critical and every score reaches an R-precision of about 0.98; here single-field retraining ranks as well as $M^{(2)}$. Single-field risk is therefore a competitive ranking baseline when most fields are critical; the advantage of $M_{i \rightarrow y}^{(K)}$ is clearest in threshold-based flagging, withholding, and ranking at the higher threshold. Attribution scores computed on one fitted model (LOCO, permutation importance, SAGE) are the weakest throughout, because they measure what that model uses rather than what an adversary can infer.

C. Accuracy and efficiency of the amortized attacker

a) *Fidelity*: Fig. 7 compares the amortized estimates with retrained ground truth for all sets of size ≤ 3 . The estimate of V is biased downward, visibly so for PJM, where the training

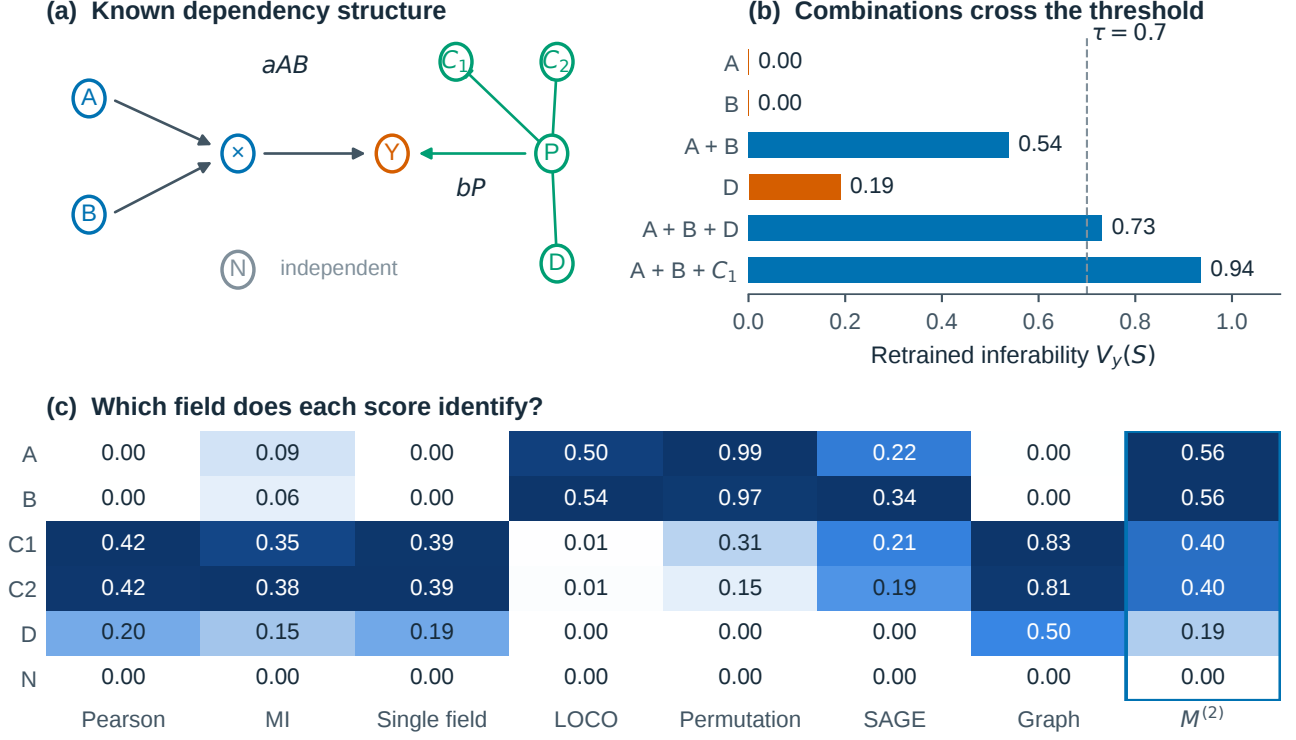


Fig. 5. Synthetic mechanism, $Y = aAB + bP + \sigma\epsilon$. (a) Dependency structure: A, B act jointly; C_1, C_2, D are noisy observations of P , and N is independent noise. The independent target-noise term is omitted from the drawing. (b) Retrained inferability of selected sets. (c) Eight field scores; numbers are raw scores and color is normalized within each method. All 63 nonempty field sets are retrained.

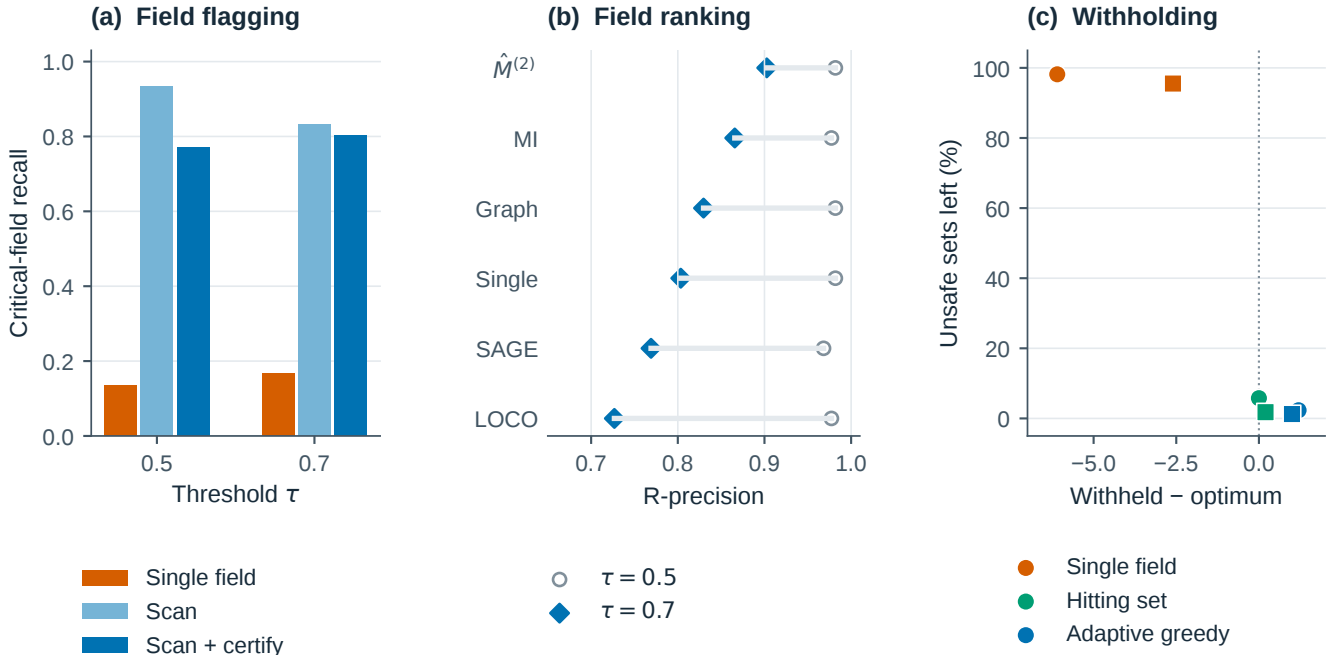


Fig. 6. Field assessment and release decisions across ten targets. (a) Mean recall; scan and scan-certify use $\delta = 0.05$. (b) Mean R-precision for six representative scores; Table IV reports all methods. (c) Mean withheld fields minus the mean optimum, against the pooled percentage of minimal unsafe sets left intact. Circles and squares indicate $\tau = 0.5$ and 0.7 , respectively. The two estimate-driven withholding strategies use $\delta = 0.05$.

TABLE III

DECISIONS OVER THE TEN TARGETS. TOP: RECALL AND PRECISION OF τ -CRITICAL FIELDS. BOTTOM: FIELDS WITHHELD (OPTIMUM IN PARENTHESES) AND SHARE OF TRUE MINIMAL UNSAFE SETS LEFT UNBROKEN. SELECTION USES ESTIMATES ONLY; RETRAINED VALUES ARE USED FOR CERTIFICATION AND EVALUATION. δ : CONSERVATIVE MARGIN OF THE SCAN.

Flagging rule	$\tau = 0.5$		$\tau = 0.7$	
	recall	precision	recall	precision
Single-field rule $V(\{i\}) > \tau$	0.13	1.00	0.17	1.00
Scan on $\hat{V}$, $\delta = 0.05$	0.93	0.99	0.83	0.80
Scan + certify, $\delta = 0$	0.88	1.00	0.75	1.00
Scan + certify, $\delta = 0.05$	0.77	1.00	0.80	1.00
Withholding strategy	withheld	left	withheld	left
Single-field grading	3.3 (9.4)	98.2%	2.5 (5.1)	95.5%
Hitting set of MUS on $\hat{V}$, $\delta = 0.05$	9.4 (9.4)	5.8%	5.3 (5.1)	1.8%
Adaptive $\hat{M}^{(2)}$ greedy, $\delta = 0.05$	10.6 (9.4)	2.3%	6.1 (5.1)	1.3%

TABLE IV

RANKING QUALITY OVER THE TEN TARGETS. R-PRECISION: SHARE OF TRUE CRITICAL FIELDS AMONG THE TOP- c FIELDS, c BEING THE TRUE NUMBER OF CRITICAL FIELDS (A THRESHOLD THAT FAVOURS THE BASELINES). EXCESS: FIELDS WITHHELD IN SCORE ORDER UNTIL NO MUS REMAINS, MINUS THE OPTIMUM.

Score	R-precision		excess withheld	
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.5$	$\tau = 0.7$
Pearson r^2	0.977	0.815	5.4	3.0
Spearman ρ^2	0.977	0.767	3.5	2.8
Mutual information	0.977	0.866	2.0	2.1
Distance correlation	0.982	0.833	3.0	2.0
Single-field $V(\{i\})$	0.982	0.803	1.6	1.9
LOCO	0.977	0.727	11.0	8.0
Permutation importance	0.968	0.795	8.9	4.7
SAGE	0.968	0.769	8.6	6.3
Inference-graph propagation	0.982	0.830	5.6	2.7
$\hat{M}^{(2)}$ (amortized, no retraining)	0.982	0.903	3.0	0.9
$M^{(2)}$ scan-certify ($k = 3$)	0.982	0.888	1.6	0.4
$M^{(2)}$ exact	0.977	0.894	1.5	0.4

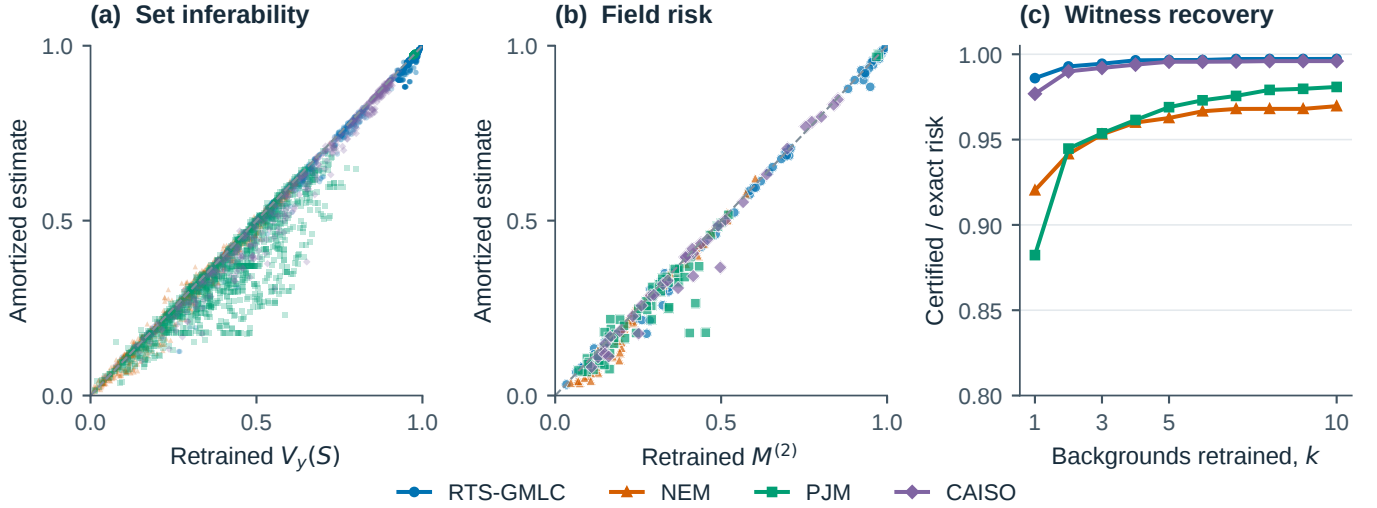


Fig. 7. Amortized estimates against retrained ground truth. (a) $\hat{V}_y(S)$ for sets of size ≤ 3 (random subsample). (b) $\hat{M}^{(2)}$ for every field and target. (c) Certified lower bound over exact $M^{(2)}$ when the top- k scanned backgrounds per field are retrained.

set is smallest; averaged over the ten targets the error is 0.026 with a signed bias of -0.024 , indicating predominantly downward errors. Consistent with the cancellation mechanism in (5), the field-level risk is estimated more accurately than the set values it is built from: the mean absolute error of $\hat{M}^{(2)}$ is 0.021 and its Spearman correlation with the exact $M^{(2)}$ across fields is 0.957 (Table V). Certification recovers most of the exact risk from very few retrained backgrounds: retraining the single best-scanned background per field yields a lower bound equal to 93.4% of the exact $M^{(2)}$ on average, three backgrounds yield 97.0%, and on RTS-GMLC and CAISO the top-3 ratio exceeds 99% (Fig. 7c). For a field set of 35 candidates this replaces the $1 + 34 + \binom{34}{2} = 596$ backgrounds per field of an exhaustive evaluation by three.

b) What matters in the estimator: Table V and Fig. 8 separate the components. The largest single effect is solving the readout per field set: with the same backbones, reading the prediction off a shared output head gives an $M^{(2)}$ error of 0.057, against 0.021 for the per-set readout, and a lower certified ratio. The second is a nonlinear feature map: without a backbone, a linear readout (0.144) or an in-set quadratic dictionary (0.077) is far less accurate. A backbone trained on fully visible inputs is less accurate (0.036, Spearman 0.928), whereas the choice of masking distribution has a smaller effect (Bernoulli versus uniform visible count: 0.024 versus 0.023). The numerical safeguards remove the overestimation of $M^{(2)}$ on real data (bias $+0.009$ without them, certified top-1 ratio 0.825 instead of 0.936). Averaging three seeds reduces the V error from 0.030 to 0.026.

A randomly initialized, untrained backbone with the same per-set readout estimates V and $M^{(2)}$ about as well as the pre-trained one (0.027 and 0.022). The per-set readout on random ReLU features of the masked input is itself a random-feature regression [43], which is already a good model of a set’s inferability. What pre-training adds is the ordering of the largest marginals, which is what certification relies on: the certified top-1 ratio is 0.936 for the pre-trained and 0.906 for the random backbone, and 0.940 versus 0.694 on PJM load, the hardest target. We therefore attribute the accuracy of the estimator primarily to the per-set readout and the conditional feature map, and the reliability of the scan to masked pre-training.

Warm-start fine-tuning of the whole network for each set, the natural fast alternative to retraining [25], is less accurate than the closed-form readout on nine of ten targets (on 150 random sets per data set; e.g. 0.028–0.044 versus 0.013–0.021 on RTS-GMLC) while being slower.

c) Extension: sharing the backbone across target definitions: In practice the auditor’s target list and candidate pool change—a field is reclassified, a new target is added. With masked reconstruction over all 53 columns of a market, one backbone serves every such definition through the readout alone. Table VI compares it with backbones supervised on each group’s own targets. The all-column backbone is more accurate for V (0.040 versus 0.049) and slightly more accurate for $M^{(2)}$ (0.026 versus 0.029), and its certified top-1 ratio rises from 0.906 to 0.944.

The mechanism is visible in the representation. A back-

bone that serves one or two targets leaves a quarter of its hidden units inactive (24%) and has an effective dimension of 1.45; one that must reconstruct every column keeps 92% of units active and has an effective dimension of 2.44. This is consistent with multi-task representation learning [34], [35], [36]: reconstructing more operational quantities encourages a broader representation of their shared variation. The larger improvement in V than in M is also consistent with (5), since a reduction in shared bias need not change marginal differences to the same extent.

The last row of Table VI tests a target never seen in pre-training: the target column is deleted from the data before pre-training. Set values and marginals are estimated as well as with the all-column backbone, but $\hat{M}^{(2)}$ is inflated (bias $+0.028$), because the maximum over many marginals picks up the few that are overestimated; the certified top-1 ratio drops to 0.75. Retraining three backgrounds per field restores it to 0.966. The recovery from three backgrounds supports using certification when extending the scan to targets unseen during pre-training.

d) Cost: Measured on one otherwise idle GPU, retraining the attacker family sequentially for one field set takes 14–56 s (the perceptrons dominate; one network per target), whereas the amortized estimate takes 62–78 ms, a factor of 191–763. A complete $K = 2$ table (all sets of size ≤ 3) therefore takes 2–9 min to scan, against 7–53 h of sequential retraining; certification adds three retrained backgrounds per field and target, i.e. a few hundred sets. Pre-training a backbone takes about half a minute and is done once per data set. The savings compound over the recurrent workloads of an audit: re-grading after every release, redefined targets, and the sets of size four needed for $K = 3$ disposal.

D. Robustness and background-budget coverage

a) Attacker seeds and data splits: The retrained reference varies with the experimental run: attackers are trained with random initializations and early-stopping splits, and V_y is measured on one random test split. We repeated the complete three-attacker retraining with two further seeds (RTS-GMLC, three targets; PJM load) and on two further random 70/30 splits (RTS-GMLC). Across these reruns the mean absolute change in V_y is 0.001–0.008, the Spearman correlation of $M^{(2)}$ across fields with the main run is 0.983–0.999, the set of τ -critical fields is identical at $\tau = 0.5$ (Jaccard 1.0 in every case) and agrees with Jaccard 0.93–1.00 at $\tau = 0.7$, and the number of combination-critical fields and the minimum withholding set change by at most one or two fields (Fig. 9c). The conclusions of Sections VI-A–VI-B are therefore not artefacts of a particular seed or split.

b) Threshold: Fig. 9(d) sweeps τ from 0.3 to 0.9 over all ten targets. At every threshold the share of fields that are safe alone but critical with at most two background fields exceeds the share of fields that are unsafe alone (e.g. 43% versus 10% at $\tau = 0.7$), and withholding the fields flagged by single-field grading leaves between 38% (at $\tau = 0.9$) and 96% (at $\tau = 0.5$) of the minimal unsafe sets intact. The gap between the two grading philosophies is thus not tied to the choice of τ .

TABLE V

ESTIMATOR COMPARISON AND ABLATION, AVERAGED OVER THE TEN TARGETS AND ALL SETS OF SIZE ≤ 3 . ABLATIONS USE ONE SEED AND SHOULD BE COMPARED WITH “OURS, ONE SEED”. CERT.@ k : CERTIFIED $L^{(2)}$ OVER EXACT $M^{(2)}$ WHEN k BACKGROUNDS PER FIELD ARE RETAINED.

Estimator	MAE V	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	Spearman	cert.@1	cert.@3
Linear readout on $x \odot m$ (no backbone)	0.212	0.068	0.144	-0.139	0.737	0.803	0.878
Readout on $[x, x^2]$ (no backbone)	0.132	0.049	0.096	-0.089	0.854	0.859	0.912
In-set quadratic dictionary (no backbone)	0.120	0.045	0.077	-0.066	0.870	0.872	0.932
Shared output head (3 seeds)	0.067	0.030	0.057	-0.054	0.930	0.883	0.939
Untrained random backbone + readout	0.027	0.017	0.022	-0.015	0.959	0.906	0.971
Pre-training without masks	0.032	0.020	0.036	-0.004	0.928	0.898	0.963
Bernoulli(0.5) masks	0.030	0.019	0.024	-0.018	0.958	0.932	0.969
Readout on ϕ only	0.030	0.019	0.025	-0.016	0.950	0.927	0.964
No numerical safeguards	0.031	0.020	0.036	+0.009	0.904	0.825	0.946
Ours, one seed	0.030	0.019	0.023	-0.019	0.953	0.936	0.967
Ours, three seeds (main)	0.026	0.017	0.021	-0.018	0.957	0.934	0.970

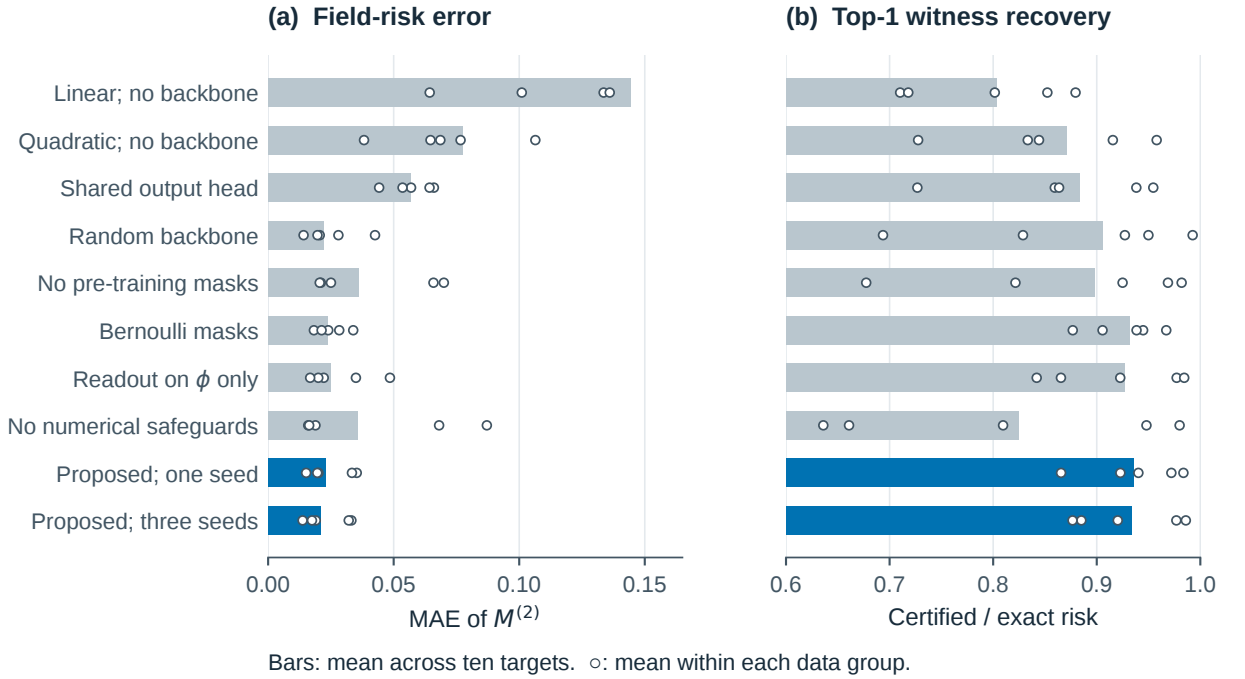


Fig. 8. What drives the estimator’s performance? (a) Mean absolute field-risk error and (b) the certified ratio from the top-ranked background. Bars are means over ten targets; open circles are means within each data group. Blue identifies the complete estimator. Set-value and ranking metrics appear in Table V; the separate backbone-scope extension appears in Table VI.

TABLE VI

BACKBONE SCOPE ON THE FOUR PJM/CAISO TARGETS (SAME READOUT). INACTIVE: HIDDEN UNITS WITH ZERO VARIANCE ON THE TRAINING ROWS; EFF. DIM: PARTICIPATION RATIO OF THE FEATURE COVARIANCE; COMMON-MODE: $|\bar{e}|/|e|$ OF THE V ERROR e .

Backbone	inactive	eff. dim	MAE V	common-mode	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	cert.@1	cert.@3
Per-group (targets supervised)	24.2%	1.45	0.049	98%	0.028	0.029	-0.025	0.906	0.963
All-column (masked reconstruction)	7.7%	2.44	0.040	94%	0.024	0.026	-0.021	0.944	0.976
All-column, target held out	7.5%	2.50	0.039	94%	0.024	0.056	+0.028	0.750	0.966

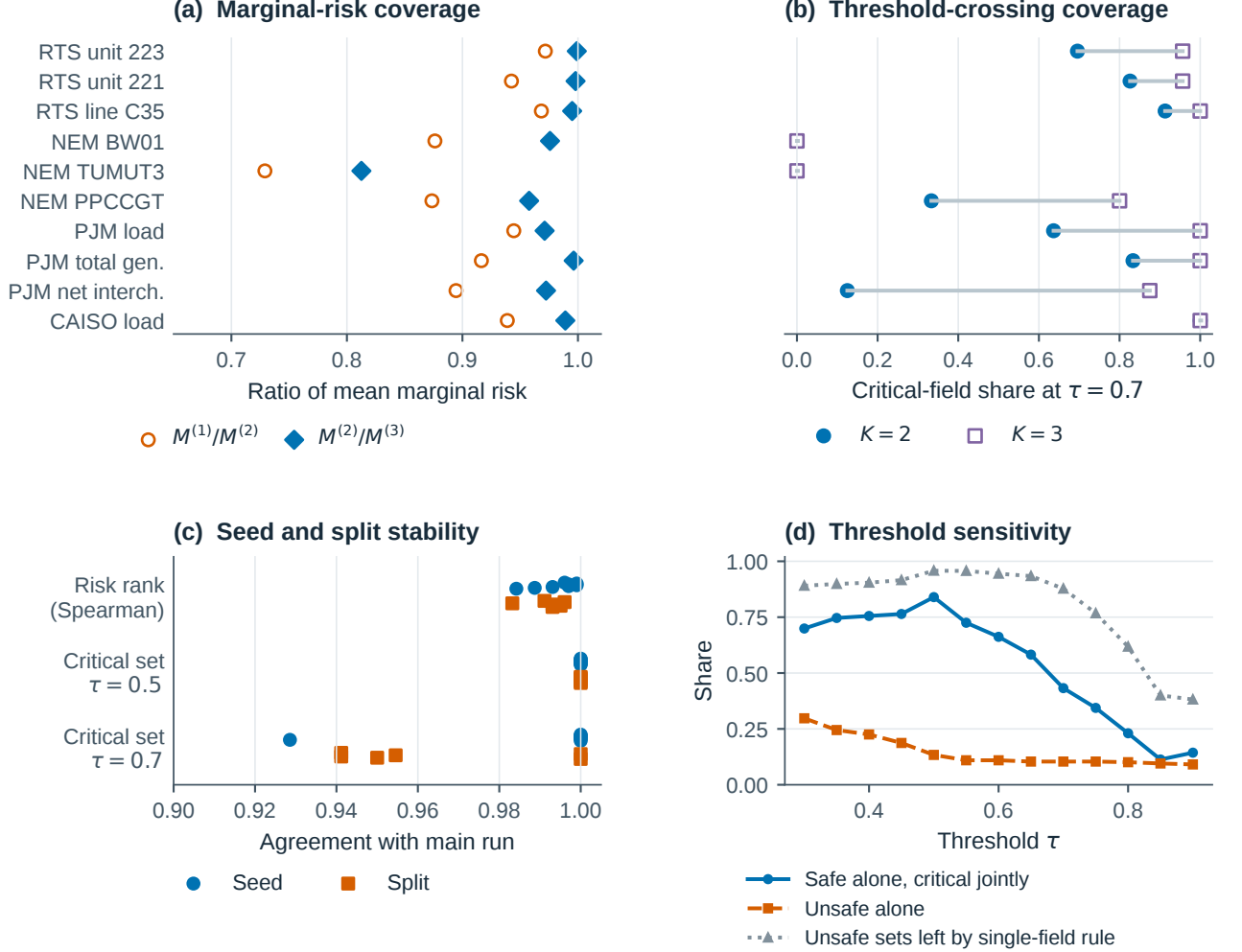


Fig. 9. Robustness. (a) Saturation of mean field risk in the background budget. (b) Share of τ -critical fields at $\tau = 0.7$ with $K = 2$ (filled) and $K = 3$ (open). (c) Agreement of reruns with different attacker seeds or data splits with the main run. (d) Threshold sweep over ten targets: fields that are safe alone but critical (solid), fields unsafe alone (dashed), and minimal unsafe sets left after single-field grading (dotted).

c) Choice of K : Exhaustive retraining of all sets of four fields on every data set (for NEM an additional 27,405 sets) gives the exact $M^{(3)}$. Measured by the *marginal* risk, $K = 2$ is close to saturation: $M^{(2)}/M^{(3)}$, the ratio of mean field risk, lies between 0.958 and 1.000 for nine targets and is 0.813 for NEM TUMUT3, a hydro unit whose dispatch depends on inter-regional exchange (Fig. 9a). The first background field brings most of the escalation ($M^{(1)}/M^{(2)}$ between 0.73 and 0.97).

Threshold crossings, however, do not saturate as quickly, as anticipated in Section III-C. At $\tau = 0.7$ several targets have fields that become critical only with a third background field: PJM net interchange (3 critical fields at $K = 2$, 21 at $K = 3$), NEM PPCCGT (10 and 24) and PJM load (14 and 22) (Fig. 9b). At $\tau = 0.5$ all fields are already critical at $K = 2$ except for TUMUT3 (22 versus 30). Minimal unsafe sets of size four therefore exist that a hitting set computed at $K = 2$ does not break. Breaking all minimal unsafe sets up to size four requires on average 11.3 withheld fields instead of 9.4 at $\tau = 0.5$, and 6.3 instead of 5.1 at $\tau = 0.7$. These

results support $K = 2$ as an economical budget for marginal-risk grading in this evaluation, with $K = 3$ providing wider coverage for withholding decisions at the cost of one to two additional withheld fields. Evaluating all sets of size four is exactly the workload for which the amortized scan is needed: for CAISO it comprises 52,360 sets per target, against 7,175 for $K = 2$.

VII. DISCUSSION

a) A grade with an explanation: The practical contribution is a field-level assessment that retains the context in which information becomes sensitive. The marginal score identifies how much a field can add; its witness identifies the accompanying fields. This distinction explains why low single-field predictability need not imply low release risk. The same set-level evaluation supports a complementary output—minimal unsafe combinations—from which withholding strategies can be constructed. In the experiments, this combination-aware view substantially reduces the unsafe combinations left intact by single-field grading.

b) *Efficient search over combinations*: The scan and the subset-specific readout address the main computational burden: evaluating many possible information sets. The ablations show that adapting the readout to each set is more consequential than increasing training effort per query. Random nonlinear features are already effective for average estimation error, while masked pre-training improves the selection of high-value backgrounds for certification. The all-column reconstruction experiment suggests a route to sharing this representation across changing target definitions. Inference graphs could likewise guide background selection by supplying physical relations [6]; the present evaluation focuses on exhaustive scans of small sets.

c) *Scope of the assessment*: The reported risks depend on the attacker family, the available labelled history, and the sampled operating conditions. Random splits evaluate held-out observations from the same data collection; temporal changes may alter the measured values. The background budget also specifies the combinations covered by the audit. Although $K = 2$ captures most measured marginal risk for nine targets, size-four combinations remain relevant to threshold-based withholding. Finally, field roles follow a common disclosure-based scenario rather than an assessment of each market’s local rules, and the inference-graph baseline evaluates statistical propagation without expert labels or the attention training of [6].

VIII. CONCLUSION

This paper connects set-dependent inference leakage to field-level risk grading of power-system data. Budgeted marginal inference measures a field’s largest predictive contribution over admissible backgrounds and identifies a witness combination. An amortized attacker makes the search practical through masked pre-training and a closed-form readout for each field set, while selected retrained backgrounds yield lower bounds on the marginal risk.

Across four data sets, single-field grading recovers only 13–17% of critical fields. Scan–certify recovers up to 88% with no false positives against the retrained reference; estimate-driven withholding leaves only 1–6% of minimal unsafe combinations intact with near-minimal withholding. The results support assessing fields through their combinations, using subset-specific adaptation to make that assessment efficient, and selecting the background budget according to whether the audit prioritizes marginal-risk ranking or coverage of unsafe releases.

DATA AND CODE AVAILABILITY

RTS-GMLC [39] and the AEMO NEMWEB archive [40] are public; the PJM and CAISO series were obtained from the operators’ public data portals. Field-role tables, dispatch simulation, retraining, estimator, and analysis code will be made available upon publication.

APPENDIX A

FURTHER PROPERTIES OF BUDGETED MARGINAL RISK

Proposition 3 (Ladder and candidate-set monotonicity). (a) $M_{i \rightarrow y}^{(0)} \leq M_{i \rightarrow y}^{(1)} \leq \dots \leq M_{i \rightarrow y}^{(|H|-1)}$; (b) *enlarging the*

candidate set from H to $H' \supseteq H$ cannot decrease $M_{i \rightarrow y}^{(K)}$; (c) if f' duplicates f , i.e. $V_y(B \cup T \cup \{f'\}) = V_y(B \cup T \cup \{f\}) = V_y(B \cup T \cup \{f, f'\})$ for all T , deleting f' leaves $M_{i \rightarrow y}^{(K)}$ unchanged for every $i \neq f, f'$.

Proof. (a) and (b) are maxima over nested feasible sets. (c) If a witness T of i contains f' , replace f' by f (if $f \notin T$) or drop it (if $f \in T$); the value is unchanged and the budget is respected, so the maximum does not decrease after deleting f' , and by (b) it does not increase. $\square$

Theorem 3 (Truncated super-efficiency). *For $A \subseteq H$ with $|A| \leq K + 1$, $V_y(B \cup A) - V_y(B) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$. More tightly, with $U^{(K)}(\emptyset) = 0$ and $U^{(K)}(A) = \min_{j \in A} [U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(\min(|A|-1, K))}]$, $V_y(B \cup A) - V_y(B) \leq U^{(K)}(A) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$.*

Proof. Order $A = \{i_1, \dots, i_m\}$ and telescope $v_B(A) = \sum_t [v_B(A_t) - v_B(A_{t-1})]$; the t -th term is a marginal of i_t over a background of size $t - 1 \leq K$, hence at most $M_{i_t \rightarrow y}^{(K)}$. The staged bound follows by induction on $|A|$ and Proposition 3(a). $\square$

For $K \geq |H| - 1$ this is the super-efficiency of MCI [7]; with a finite budget it is stronger because $M_{i \rightarrow y}^{(K)} \leq \text{MCI}$. It yields a release rule “accept A if $U^{(K)}(A) \leq \tau$ ” that never accepts an unsafe set of size $\leq K + 1$. We do not use this rule in the main text because it is conservative for larger release sets; the minimum-hitting-set withholding of Section VI-B addresses the same need directly.

Proposition 4 (Irreducible synergy is bounded by field risk). *With $\text{syn}^B(S) = V_y(B \cup S) - \max_{T \subsetneq S} V_y(B \cup T)$ and $|S| \geq 2$, $\text{syn}^B(S) \leq \min_{j \in S} M_{j \rightarrow y}^{(|S|-1)}$.*

Proof. $\max_{T \subsetneq S} V_y(B \cup T) \geq V_y(B \cup S \setminus \{j\})$ for every j , hence $\text{syn}^B(S) \leq \Delta_j(B \cup S \setminus \{j\}) \leq M_{j \rightarrow y}^{(|S|-1)}$. $\square$

Remark 1 (Escalation is weaker than synergy). $M_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(0)}$ iff $\Delta_i(B \cup T) \leq \Delta_i(B)$ for all $|T| \leq K$. Submodularity implies no escalation but not conversely: with $V_y(\{1\}) = 0.8$, $V_y(\{2\}) = 0.6$, $V_y(\{3\}) = 0.8$, all pairs 0.8 and $V_y(\{1, 2, 3\}) = 1$, every field has $M^{(2)} = M^{(0)}$, yet $\Delta_1(\{3\}) = 0 < \Delta_1(\{2, 3\}) = 0.2$.

APPENDIX B

COMPARISON OF FIELD-RISK SEMANTICS

Table VII summarizes six criteria for the release setting: attack-based evaluation (R1), joint inference (R2), worst-case contexts (R3), resistance to dilution by substitutes (R4), a finite background budget (R5), and an explicit public baseline (R6). The budget and baseline specialize marginal contribution to the auditor’s admissible information sets.

APPENDIX C

IMPLEMENTATION DETAILS

a) *Attackers*: Single- and multi-target perceptrons: two hidden layers of 128 ReLU units, dropout 0.15, Adam (learning rate 10^{-3} , weight decay 5×10^{-4}), batch 128, at most

TABLE VII
REQUIREMENTS FOR A FIELD-LEVEL INFERENCE-RISK QUANTITY. ✓: SATISFIED; ○: PARTIALLY; −: NOT SATISFIED.

Method	R1 checkable	R2 joint	R3 worst case	R4 no dilution	R5 bounded K	R6 baseline B
Correlation / mutual information	−	−	−	✓	−	−
Single-field retraining $M^{(0)}$	✓	−	−	✓	−	○
Inference-graph propagation [6]	−	○	−	○	−	−
Shapley / SAGE [15]	✓	✓	−	−	−	−
LOCO / permutation	✓	✓	−	−	−	−
MCI [7]	✓	✓	✓	✓	−	−
$M_{i \rightarrow y}^{(K)}$ (this paper)	✓	✓	✓	✓	✓	✓

300 epochs, patience 60, epoch chosen per (set, target) on validation. Many attackers are trained in parallel on one GPU as independent networks whose inputs are masked to their own field set; masked columns receive no gradient, so each network is functionally a network trained on its set alone. Gradient boosting: histogram gradient boosting with (learning rate, leaves) $\in \{(0.05, 15), (0.1, 31)\}$, 300 iterations with internal early stopping; the configuration is chosen on validation. The family member is chosen per (set, target) on validation. Across data sets gradient boosting is selected for 35–76% of (set, target) pairs (RTS-GMLC 76%, NEM 35%, PJM 67–69%, CAISO 52%), so the family maximum matters.

b) *Estimator*: Backbone: three hidden layers of 256 ReLU units with dropout 0.1, input $[x \odot m, m]$, Adam (10^{-3} , weight decay 5×10^{-4}), batch 256, at most 400 epochs, patience 60 on eight fixed validation masks. Ridge penalties $\lambda \in \{10^{-3}, \dots, 10^2\}$.

c) *Inference-graph baseline*: Following the structure of [6] without its expert-assigned intrinsic scores: nodes are the candidate fields and the target; the weight of edge $i \rightarrow j$ is the validation R^2 of predicting field j from field i alone by gradient boosting; the score of i is $\sum_{h=1}^3 0.5^{h-1} (W^h)_{i,y}$. The first term equals single-field predictability; longer paths add inference chains through other fields.

d) *SAGE*: Marginal imputation with 32 background rows, 512 evaluation rows from the validation split and 24 random permutations, on a gradient-boosting model fitted to all candidates.

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TABLE VIII
FIELD ROLES AND RULE BASIS (CHINESE ELECTRICITY-MARKET DISCLOSURE RULES [1]; APPENDIX ITEM NUMBERS IN PARENTHESES).

Data set	Candidate fields (released before the target)	Targets
RTS-GMLC	zonal load forecasts (3; 6.41); wind, PV, rooftop PV and hydro forecasts (4; 6.42); regulation up/down and spinning reserve requirements (3; 6.46); system energy price and nodal prices at 9 buses (10; 6.52); congestion indicators of lines C6, C29, CA-1 (3; 6.54)	outputs of units 223_STEAM_3 and 221_CC_1 (specific information, Art. 17(3)); hourly flow on line C35 (important-line disclosure category, 6.66)
NEM 2024	regional demand (5), available capacity (5), semi-scheduled wind and solar forecasts (9), regional prices (5), interconnector flows (6)	actual outputs of units BW01, TUMUT3, PPCCGT (specific information)
PJM	regulation market prices and volumes (5); day-ahead reserve prices, requirements, self-schedules and cleared volumes (9); day-ahead system energy price, LMP, congestion and loss components (4); real-time LMP, congestion and loss components (3); scheduled interchange (1); for total generation and net interchange also the system load forecasts (2)	metered load (6.61), total generation (6.57), net actual interchange (6.65), all released next day
CAISO	day-ahead load forecasts of three sub-areas (3); day-ahead wind and solar forecasts (6); day-ahead and 15-min real-time prices and components (11); day-ahead schedules of sub-area load, total generation, imports and exports (6); ancillary-service requirements incl. regulation mileage (5) and procured volumes (4)	actual CAISO load (6.61), released next day

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